

Progress Report: FinVerify – Improving the Reliability of LLM-based Financial Advisors Through Hybrid Verification Systems

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1. Project Name

FinVerify: A Hybrid Verification System for LLM-Based Financial Advisors

2. Overview

The goal of this project is to build a hybrid pipeline for LLM-based financial advice that adds a verification layer between the raw LLM output and the user. During the midterm literature review, I found that LLM-based systems represent a paradigm shift for digital financial advisors due to their conversational ability and personalization, but that hallucinations pose serious risks in high-stakes financial contexts. This project addresses that gap directly.

The system works as follows: when a user asks a financial question, Claude Sonnet (via the Anthropic API) generates a general response. Before that response reaches the user, a verification layer checks it against trusted sources, including financial literacy datasets, regulatory documents, and predefined financial rules. This will be done using RAG retrieval with a vector store of trusted financial documents, combined with a second LLM call that receives both the original response and the retrieved context and is prompted to identify factual inconsistencies. I chose this approach for its feasibility within the project timeline: RAG retrieval is straightforward to implement using a framework like LangChain with a FAISS or Chroma vector store. If the verifier detects an inaccuracy, it either corrects the output using retrieved sources or flags it and requests a regeneration.

The project compares this hybrid system against a baseline LLM-only system across a dataset of personal finance questions, measuring answer accuracy, hallucination rate, and error detection rate. Since the original project proposal in March, I've finalized the system architecture

(specifically the choice of Claude Sonnet as the primary LLM and RAG-based retrieval + LLM-as-judge for the verification layer) and started working on integrating this into a GitHub repository.

3. Research Questions

- RQ1:** How frequently do LLM-based financial advice systems hallucinate or produce incorrect information when answering common and advanced financial questions?
- RQ2:** Can a verification layer that references structured financial rules and trusted documents improve response accuracy compared to a baseline LLM-only system?
- RQ3:** What types of financial questions benefit most from verification (ex: factual/regulatory questions vs. strategic/subjective advice)?

4. Value to User Community

The prospective users of this project fall into three groups:

Developers building financial AI products.

The verification layer as a concept in this project is designed to be extendable work. By open-sourcing the RAG-based verification pipeline and the prompts used for the LLM-as-judge component, developers who are building their own financial chatbots can integrate the same architecture into their systems without starting from scratch. The architecture is also not tied to a specific LLM API provider, as the Claude Sonnet model could be swapped out for GPT, DeepSeek, Grok, etc.

Researchers studying hallucination mitigation.

The evaluation benchmark I'll be creating and using, including the curated financial question dataset, scoring framework, and baseline vs. hybrid model comparison results, contributes towards the broader study of hallucinations in LLMs. Specifically, I hope that this can be used for future research into identifying and mitigating hallucinations in financial contexts for LLMs.

End users seeking trustworthy financial guidance.

Ultimately, I am working on this project to help young adults and individuals who would interact with LLM-based financial advisors. While this project does not deploy a user-facing product, it demonstrates and measures a mechanism for making such products safer, more reliable, and something that can help these groups better manage their finances.

5. Demo Plan

Elevator pitch (1 min):

With the emergence of LLMs, young adults and other users have turned towards them as financial advisors, however, they can hallucinate. In fact, our baseline testing shows that a standalone LLM gives incorrect or misleading financial advice roughly X (TBD from experiments)% of the time. FinVerify adds a verification layer that cross-checks every response against trusted financial sources before it reaches the user, reducing the error rate to Y%. The key insight of this is that you don't have to make the LLM itself better, you just need to catch its mistakes before they cause harm.

Demo (5 min):

For the demo, I'll be presenting a recorded video walkthrough of the experiments. They will show:

- A side-by-side comparison of the LLM-only baseline and the hybrid system answering the same financial question, showing how the outputs differ.
- A walkthrough of a specific example where the verification layer detects a hallucination, retrieves the relevant trusted source, and corrects the response before it would reach the user.
- Summary metrics across the full evaluation dataset (accuracy, hallucination rate, detection rate).

6. Delivery

The project will be delivered as a public GitHub repository containing:

- The evaluation dataset (or scripts to generate/assemble it; external datasets referenced by URL, not copied in).
- The verification layer implementation.
- The evaluation framework and scoring scripts.
- Results and analysis.
- A final written report.